# In class lab #9 for Numerical Analysis

**Your name:** _____

**Submit both your code and results before 23:59 Thur, May 9 or you will get 0.**

**Euler's Method**

Although any textbook on the subject of ordinary differential equations details a number of methods for explicitly finding solutions to first-order initial-value problems, in practice few of the problems originating from the study of physical phenomena can be solved exactly. So let us learn the numerical method to solve ordinary differential equations,

The first method we will study is Euler's Method for an initial value problem for a first-order ordinary differential equation,

$$\begin{cases} y' = f(t, y) \\ y(a) = y_a \\ t \text{ in } [a, b]. \end{cases} \tag{1}$$

We begin with a grid of $n + 1$ points

$$t_0 < t_1 < t_2 < \cdots < t_n$$

along with the $t$-axis with equal step size $h$.

Begin with $w_0 = y_0$. Following the slope field at each $t_i$ yields the approximation

$$w_{i+1} = w_i + hf(t_i, w_i) \tag{2}$$

at $t_{i+1}$, since $f(t_i, w_i)$ represents the slope of the solution (illustrate it from the points of view of numerical differentiation and numerical integration ). Note that the change in $y$ is the horizontal distance $h$ multiplied by the slope.

The formula for **Forward Euler's Method** can expressed as follows:

$$w_0 = y_0 \tag{3}$$
$$w_{i+1} = w_i + hf(t_i, w_i) \tag{4}$$

The formula for **Backward Euler's Method** can expressed as follows:

$$w_0 = y_0 \tag{5}$$
$$w_{i+1} = w_i + hf(t_{i+1}, w_{i+1}) \tag{6}$$

**Problem 1.** (20 points) Consider the initial value problem of the following differential equation

$$\begin{cases} y' = t \\ y(0) = 2. \end{cases} \tag{7}$$

(a) Use separation of variables to find the solutions of equation (7).

(b) Apply Forward Euler's Method with step size $h = 1/4$ to the equation (7) on the interval $[0, 1]$. List the $w_i, i = 0, \cdots, 4$, and find the error at $t = 1$ by comparing with the correct solution.

**Problem 2.** (40 points) Use Euler's method to approximate the solution for each of the following initial-value problems. Output time steps t and the numerical solution y.

a) $y' = \left(\frac{y}{t}\right)^2 + \frac{y}{t}, 1 \le t \le 1.2, \quad y(1) = 1$ with $h = 0.1$. Forward Euler's method

c) $y' = t^3 - \frac{y}{t}, \ 1 \le t \le 2, \quad y(1) = 0.4$ with $\ h = 0.1$. Backward Euler's method

**Problem 3.** (20 points) Plot the Forward Euler's Method approximate solution on [0,1] for the differential equation $y' = 1 + y^2$ and initial condition (a) $y_0 = 0$ (b) $y_0 = 1$, along with the exact solution (a) $y = \tan(t)$, (b) $y = \tan(t + \pi/4)$. Use step sizes $h = 0.1$ and $h = 0.05$.

**Trapezoid Rule and Simpson's Rule for Numerical Integration**

**Problem 4.** (20 points) Use Midpoint Rule, Trapezoid Rule, Simpson's Rule to approximate these problems in the table below.

| $f(x)$ | $\frac{1}{x+1}$ | $\sqrt{1+x^2}$ | $\sin(x)$ | $e^x$ |
|---|---|---|---|---|
| $I(f) = \int_0^2 f(x)dx$ | 1.098612288668110 | | | |
| $M_1(f) = 2f(1)$ | | | | |
| $\lvert M_1(f) - I(f)\rvert$ | | | | |
| $T_1(f) = f(0) + f(2)$ | | | | |
| $\lvert T_1(f) - I(f)\rvert$ | | | | |
| $S_2(f) = \frac{1}{3}[f(0) + 4f(1) + f(2)]$ | | | | |
| $\lvert S_2(f) - I(f)\rvert$ | | | | |

Example for solving $I(f) = \int_0^2 f(x)dx$, where $f(x) = \frac{1}{x+1}$ by MATLAB.

```
>> syms x
>> f = 1/(x+1)

f =

1/(x + 1)

>> int(f,[0,2])

ans =

log(3)

>> format long
>> log(3)

ans =

1.098612288668110
```